

Probability-Flow Distillation: Distribution Matching in Parameter Space

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Project Page: <https://rrohithiam.github.io/projects/pfd>

- 1 Introduction
- 2 A Brief Primer
- 3 Analysis of SDS and SDI
- 4 Probability-Flow Distillation
- 5 Experiments

Outline

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Sampling in Parameter Space

- Diffusion models [1, 2, 3] sample remarkably well from complex data distributions p_0 .
- In many applications, however, the quantity of interest is not the data itself, but the **parameters** θ that generate it through a differentiable forward model

$$x_0 = g(\theta, \omega), \quad \omega \sim \xi \quad (\text{randomness of the forward model})$$

- **Goal:** find a distribution π over θ such that the pushforward matches the data,

$$\pi^* \in \arg \min_{\pi} D_{\text{KL}}(g_{\#}(\pi \otimes \xi) \parallel p_0)$$

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Examples

- **Text-to-3D** [4]: θ a 3D representation, g a renderer, ω the camera pose.
- **Audio** [5]: θ the parameters of an impact synthesizer.
- **One-step generators** [6]: $g(\theta, \omega) = G_{\theta}(\omega)$, $\omega \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, target $\pi = \delta_{\theta^*}$.

Motivating Example: Text-to-3D

- Replicating the success of image diffusion in 3D is hard due to the scarcity of high-quality 3D data.
- Stable Diffusion [3] is trained on LAION-5B, ~ 5.85 billion image-text pairs [7], whereas Objaverse contains only a few million, largely synthetic, 3D assets [8].
- Score distillation [4] sidesteps this: optimize a 3D asset so that its renders look like samples from a pretrained 2D diffusion model.



Figure: Representative samples from the Objaverse dataset [8]

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Which distribution over θ does score distillation actually induce?



Figure: Representative samples from the Objaverse dataset [8]

A Unified View of Score Distillation

Existing distillation methods reduce, directly or approximately, to one of three: **SDS** [4], **SDI** [9], and **VSD** [10]. All take the form

$$\frac{d\boldsymbol{\theta}}{d\tau} = -\mathbb{E}_{t,\omega,\epsilon} \left[\left(\frac{\partial \mathbf{x}_0}{\partial \boldsymbol{\theta}} \right)^\top \boldsymbol{\Delta}(\mathbf{x}_0, t, \epsilon) \right]$$

- $\boldsymbol{\Delta}$: gradient estimator of the method; $\partial \mathbf{x}_0 / \partial \boldsymbol{\theta}$ carries it from data space to parameter space.
- τ : optimization time, distinct from the diffusion time $t \sim \mathcal{U}[0, 1]$.
- View this as dynamics on particles $\{\boldsymbol{\theta}_i\}_i$, with induced distribution $q_{0,\tau} = \mathbf{g}_\#(\pi_\tau \otimes \xi)$ on $\mathbf{x}_0$.

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Question

VSD is derived so that $q_{0,\tau} \rightarrow p_0$. For SDS and SDI, **which distribution does $q_{0,\tau}$ converge to?**

- ➊ **Analysis.** We show that for SDS, $q_{0,\tau}$ collapses onto the modes of p_0 , and for SDI, it converges to a contracted version of p_0 . We also explain why SDI needs a *negative* CFG scale.
- ➋ **Observation.** Tweedie's posterior mean is a single Euler step of the reverse PF-ODE $\Rightarrow$ a dual-bridge extension of SDI with p_0 as a fixed point.
- ➌ **Method.** Dropping one Jacobian gives **Probability-Flow Distillation (PFD)**, which needs only the forward PF-ODE and provably beats VSD in the small-time regime.
- ➍ **Experiments.** Synthetic targets, image tasks on CelebA, and text-to-3D generation validate the analysis and effectiveness of PFD.

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The Forward Diffusion Process

Consider the forward diffusion process

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t) dt + g(t) d\mathbf{w}_t$$

where,

- $\mathbf{f}(\mathbf{x}_t, t)$: Deterministic drift term
- $g(t)$: Scalar diffusion coefficient
- $d\mathbf{w}_t$: Brownian motion increment

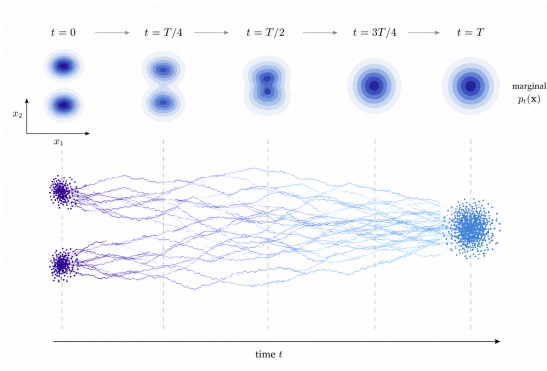


Illustration: Stochastic trajectories evolving a bimodal distribution to a unimodal Gaussian.

Reversing Diffusion with a Diffusion

Anderson [11] showed that the reverse-time dynamics of the forward diffusion process are given by

$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)] dt + g(t) d\bar{\mathbf{w}}_t$$

where,

- $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)$: Score function
- $d\bar{\mathbf{w}}_t$: Reverse-time Brownian motion

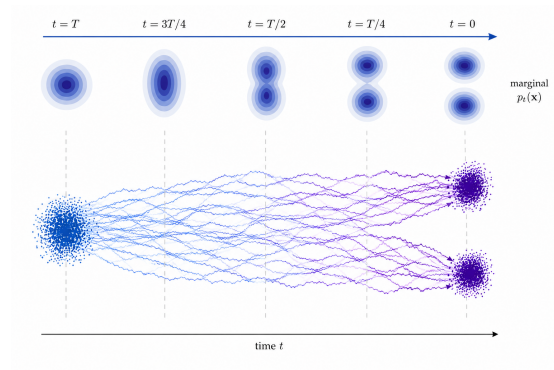


Illustration: Reverse-time stochastic trajectories transforming a Gaussian prior into a bimodal data distribution.

From Stochastic Dynamics to Deterministic Flows

Song et al. [2] showed that the reverse SDE admits an equivalent deterministic ODE

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, t) - \frac{1}{2}g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)$$

- No stochastic Brownian term. Every sample follows a deterministic trajectory.
- Same marginal densities $p_t(\mathbf{x}_t)$ as the reverse SDE.

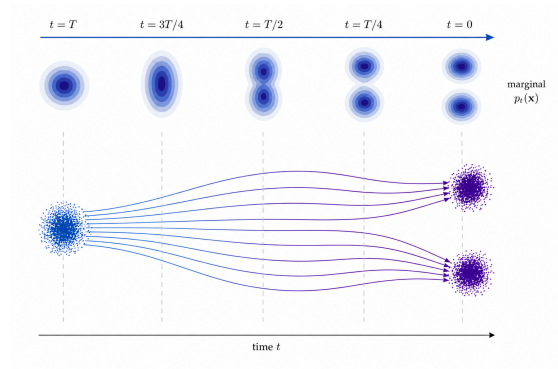


Illustration: Deterministic trajectories transporting a Gaussian distribution to the data distribution.

VE and VP Diffusion Models

The score is learned by denoising score matching [12]:

$$\mathcal{L}_{\text{DSM}}(\phi) = \mathbb{E}_{\mathbf{x}_0, \epsilon, t} [w(t) \|\epsilon - \epsilon_\phi(\mathbf{x}_t, t)\|^2]$$

With a linear drift, the perturbation kernel $p(\mathbf{x}_t | \mathbf{x}_0)$ is Gaussian, so $\mathbf{x}_t$ is sampled directly from $\mathbf{x}_0$:

Variance-Exploding (VE)

$$\begin{aligned} d\mathbf{x}_t &= \sqrt{d\sigma_t^2/dt} d\mathbf{w}_t \\ \mathbf{x}_t &= \mathbf{x}_0 + \sigma_t \epsilon, \\ \nabla_{\mathbf{x}} \log p_t &= -\frac{\epsilon_\phi(\mathbf{x}_t, t)}{\sigma_t} \end{aligned}$$

Variance-Preserving (VP)

$$\begin{aligned} d\mathbf{x}_t &= -\frac{1}{2}\beta_t \mathbf{x}_t dt + \sqrt{\beta_t} d\mathbf{w}_t \\ \mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \epsilon, \\ \nabla_{\mathbf{x}} \log p_t &= -\frac{\epsilon_\phi(\mathbf{x}_t, t)}{\sqrt{1 - \alpha_t}} \end{aligned}$$

$$\text{where } \alpha_t = \exp\left(-\int_0^t \beta_s ds\right).$$

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The PF-ODE in VE Form

For the VE-SDE, the PF-ODE is simply

$$\frac{d\mathbf{x}_t}{dt} = \boldsymbol{\epsilon}_\phi(\mathbf{x}_t, t) \frac{d\sigma_t}{dt}$$

The VP case reduces to the same form under the change of variables

$$\tilde{\sigma}_t := \sqrt{\frac{1 - \alpha_t}{\alpha_t}}, \quad \tilde{\mathbf{x}}_t := \frac{\mathbf{x}_t}{\sqrt{\alpha_t}} \implies \tilde{\mathbf{x}}_t = \mathbf{x}_0 + \tilde{\sigma}_t \boldsymbol{\epsilon},$$

$$\frac{d\tilde{\mathbf{x}}_t}{dt} = \boldsymbol{\epsilon}_\phi(\mathbf{x}_t, t) \frac{d\tilde{\sigma}_t}{dt}, \quad \mathbf{x}_t = \frac{\tilde{\mathbf{x}}_t}{\sqrt{1 + \tilde{\sigma}_t^2}}$$

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DDIM-ODE

This is exactly the DDIM-ODE [13], so we work in the VE parameterization throughout. Integrating it from $0 \rightarrow t$ maps $\mathbf{x}_0$ uniquely to $\mathbf{x}_t$: **DDIM inversion**.

CFG [14] randomly drops the condition c during training and, at inference, extrapolates

$$\hat{\epsilon}_{\phi}(\mathbf{x}_t, t, c) = \epsilon_{\phi}(\mathbf{x}_t, t, \emptyset) + \gamma (\epsilon_{\phi}(\mathbf{x}_t, t, c) - \epsilon_{\phi}(\mathbf{x}_t, t, \emptyset))$$

Since each predictor is a scaled score, $\hat{\epsilon}_{\phi} = -\sigma_t \nabla_{\mathbf{x}} \log \tilde{p}_t(\mathbf{x}_t | c)$ with the **tilted** distribution

$$\tilde{p}_t(\mathbf{x}_t | c) \propto p_t(\mathbf{x}_t | c) \left(\frac{p_t(\mathbf{x}_t | c)}{p_t(\mathbf{x}_t | \emptyset)} \right)^{\gamma-1}$$

- Sampling is driven **away from** $p_t(\cdot | \emptyset)$ and **towards** $p_t(\cdot | c)$.
- $\gamma = 1$ recovers the conditional model; $\gamma > 1$ trades diversity for alignment with c .

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Keep in Mind

The direction of this tilt matters. It will explain the negative CFG scale used by SDI.

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Score Distillation Sampling (SDS)

Idea: $g(\theta, \omega)$ should be likely under the pretrained model for every ω . Minimize the DSM loss over θ with ϕ fixed:

$$\nabla_{\theta} \mathcal{L}_{\text{DSM}} \propto \mathbb{E}_{t, \omega, \epsilon} \left[w(t) \left(\frac{\partial \mathbf{x}_0}{\partial \theta} \right)^{\top} \underbrace{\left(\frac{\partial \epsilon_{\phi}(\mathbf{x}_t, t)}{\partial \mathbf{x}_t} \right)^{\top}}_{\text{omitted in SDS}} (\epsilon_{\phi}(\mathbf{x}_t, t) - \epsilon) \right]$$

With $w(t) = \sigma_t$, the SDS estimator becomes

$$\Delta = \mathbf{x}_0 - \hat{\mathbf{x}}_0(\mathbf{x}_t), \quad \mathbf{x}_t = \mathbf{x}_0 + \sigma_t \epsilon,$$

where $\hat{\mathbf{x}}_0$ is Tweedie's posterior mean,

$$\hat{\mathbf{x}}_0(\mathbf{x}_t) = \mathbf{x}_t - \sigma_t \epsilon_{\phi}(\mathbf{x}_t, t) = \mathbb{E}_p[\mathbf{x}_0 \mid \mathbf{x}_t].$$

For the analysis, take $g = \text{id}$, so $\mathbf{x}_0 = \theta$ and $q_{0, \tau} = \pi_{\tau}$.

SDS Collapses onto the Modes

- p_0 is not a fixed point. By the law of total covariance,

$$\text{Cov}(\hat{\mathbf{x}}_0) = \text{Cov}(\mathbf{x}_0) - \underbrace{\mathbb{E}[\text{Cov}(\mathbf{x}_0 \mid \mathbf{x}_t)]}_{\succ 0},$$

so SDS pulls p_0 toward the same mean but a smaller covariance.

- **Dynamics.** Each particle evolves as

$$\frac{d\mathbf{x}_0}{d\tau} = \mathbb{E}_{t,\epsilon}[\sigma_t^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_0 + \sigma_t \epsilon)]$$

No term depends on $q_{0,\tau} \Rightarrow$ nothing maintains entropy; each particle hill-climbs to a local maximum on its own.

SDS on a Gaussian Target

Proposition 1

Let $p_0 = \mathcal{N}(\boldsymbol{\mu}, \rho^2 \mathbf{I}_d)$ and take the VE schedule $\sigma_t = t$. Then SDS reduces to the linear ODE

$$\frac{d\mathbf{x}_0}{d\tau} = -A(\rho) (\mathbf{x}_0 - \boldsymbol{\mu}), \quad A(\rho) := \mathbb{E}_t \left[\frac{t^2}{\rho^2 + t^2} \right] = 1 - \rho \arctan \frac{1}{\rho} > 0.$$

Starting from $q_{0,0} = \mathcal{N}(\mathbf{m}_0, \nu_0^2 \mathbf{I}_d)$, the particles stay Gaussian, $q_{0,\tau} = \mathcal{N}(\mathbf{m}_\tau, \nu_\tau^2 \mathbf{I}_d)$, with

$$\mathbf{m}_\tau = \boldsymbol{\mu} + e^{-A(\rho)\tau} (\mathbf{m}_0 - \boldsymbol{\mu}), \quad \nu_\tau^2 = \nu_0^2 e^{-2A(\rho)\tau}.$$

Takeaway

The mean is correct, but the variance decays to zero: $q_{0,\tau} \rightarrow \delta_{\boldsymbol{\mu}}$. **SDS collapses onto the mode.**

Score Distillation via Inversion (SDI)

- SDI [9] keeps $\Delta = x_0 - \hat{x}_0(x_t)$, but replaces the random ϵ with the noise implied by **DDIM inversion**.
- **Ideal version:** map x_0 through the PF-ODE of $\{q_{t,\tau}\}_t$ (the *forward* PF-ODE),

$$x_t = \Phi_{0 \rightarrow t}^{q_\tau}(x_0).$$

Same marginal as Gaussian perturbation, but now driven by $\nabla_x \log q_{t,\tau}$, so it **depends on** $q_{0,\tau}$.

- DDIM inversion approximates it with $\nabla_x \log p_t(\cdot \mid \emptyset) \approx \nabla_x \log q_{t,\tau}$: reasonable in the text-t0-3D setting, where the renderings x_0 of a 3D asset stay near the natural-image manifold.

Why SDI Needs a Negative CFG Scale

In practice, SDI inverts under the *target* prompt with CFG. Using $\nabla \log p_t(\cdot \mid \emptyset) \approx \nabla \log q_{t,\tau}$ in the CFG tilt (Slide 14):

$$\nabla_{\mathbf{x}} \log \tilde{p}_t(\mathbf{x}_t \mid c) \approx \nabla_{\mathbf{x}} \log \left[p_t(\mathbf{x}_t) \left(\frac{p_t(\mathbf{x}_t)}{q_{t,\tau}(\mathbf{x}_t)} \right)^{\gamma-1} \right]$$

This tilts the **reverse** PF-ODE from $q_{t,\tau}$ toward p_t . Inversion integrates the **forward** PF-ODE, so the tilt must be reversed (swap p_t and $q_{t,\tau}$):

$$\epsilon_\phi(\cdot, c) + \gamma (\epsilon_\phi(\cdot, \emptyset) - \epsilon_\phi(\cdot, c)) = \epsilon_\phi(\cdot, \emptyset) + (1 - \gamma) (\epsilon_\phi(\cdot, c) - \epsilon_\phi(\cdot, \emptyset))$$

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Negative CFG

An effective scale $\gamma > 1$ in inversion corresponds to a user-specified scale $\gamma_{\text{cfg}} = 1 - \gamma < 0$. $\gamma_{\text{cfg}} = 0$ recovers unguided inversion.

Sanity Check: Inversion-Based Image Editing

Invert, then sample under the target prompt:

- 1 Source prompt, $\gamma_{\text{inv}} = +3.5$: standard, but unfaithful under guidance.
- 2 Target prompt, $\gamma_{\text{inv}} = +3.5$: tilts the wrong way $\Rightarrow$ **unstable edits**.
- 3 Target prompt, $\gamma_{\text{inv}} = -3.5$ (as in SDI): approximates the forward PF-ODE $\Rightarrow$ an approximate DDIB [15]; **stability restored**.

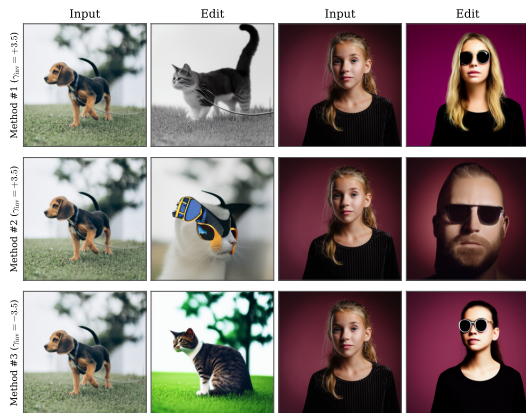


Figure: Editing under the three inversion methods (rows 1–3), with Stable Diffusion v1.5.

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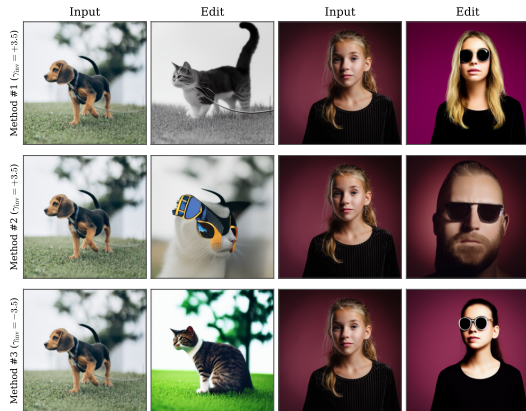


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Invert, then sample under the target prompt:

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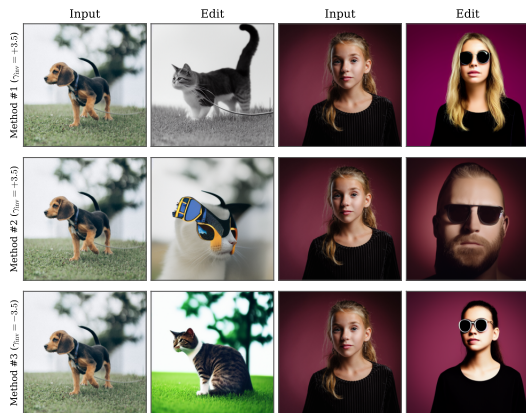


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SDI Converges to a Contracted Target

Under ideal SDI,

$$\frac{d\mathbf{x}_0}{d\tau} = \mathbb{E}_t[\hat{\mathbf{x}}_0(\Phi_{0 \rightarrow t}^{q_\tau}(\mathbf{x}_0))] - \mathbf{x}_0$$

- The dependence on $q_{0,\tau}$ acts as an **entropy term**: the more concentrated the particles, the harder the forward PF-ODE pushes them apart $\Rightarrow$ no collapse.
- But the posterior mean still contracts the spread $\Rightarrow p_0$ is still not a fixed point.

Proposition 2

Let $p_0 = \mathcal{N}(\boldsymbol{\mu}, \rho^2 \mathbf{I}_d)$ and $\sigma_t = t$. Both $\Phi_{0 \rightarrow t}^{q_\tau}$ and $\hat{\mathbf{x}}_0$ are affine, so $q_{0,\tau} = \mathcal{N}(\mathbf{m}_\tau, \nu_\tau^2 \mathbf{I}_d)$ for all τ , with

$$\frac{d\mathbf{m}_\tau}{d\tau} = -A(\rho) (\mathbf{m}_\tau - \boldsymbol{\mu}), \quad \frac{d\nu_\tau}{d\tau} = (B(\nu_\tau) - 1) \nu_\tau,$$

$$B(\nu) := \frac{\rho^2}{\nu} \mathbb{E}_t \left[\frac{\sqrt{\nu^2 + t^2}}{\rho^2 + t^2} \right],$$

and $A(\rho)$ as in Proposition 1. B is strictly decreasing, so $B(\nu_\star) = 1$ has a unique, globally attracting solution, and it satisfies $\nu_\star < \rho$.

Takeaway

$q_{0,\tau} \rightarrow \mathcal{N}(\boldsymbol{\mu}, \nu_\star^2 \mathbf{I}_d)$: no collapse, but a **strictly smaller variance** than p_0 .

Outline

- 1 Introduction
- 2 A Brief Primer
- 3 Analysis of SDS and SDI
- 4 Probability-Flow Distillation**
- 5 Experiments

Posterior Mean as a Single Euler Step

In terms of the noise level σ , the VE PF-ODE (Slide 13) reads

$$\frac{d\mathbf{x}}{d\sigma} = \boldsymbol{\epsilon}_\phi(\mathbf{x}, t(\sigma))$$

To go from $\mathbf{x}_t$ at noise level σ_t back to $\sigma = 0$, take a single explicit Euler step:

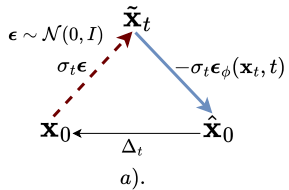
$$\mathbf{x}(0) \approx \mathbf{x}_t + (0 - \sigma_t) \boldsymbol{\epsilon}_\phi(\mathbf{x}_t, t) = \mathbf{x}_t - \sigma_t \boldsymbol{\epsilon}_\phi(\mathbf{x}_t, t)$$

Key Observation

$$\hat{\mathbf{x}}_0(\mathbf{x}_t) = \mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t] = \mathbf{x}_t - \sigma_t \boldsymbol{\epsilon}_\phi(\mathbf{x}_t, t)$$

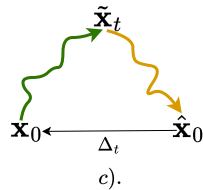
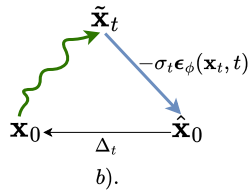
Tweedie's posterior mean is **exactly one Euler step** of the reverse PF-ODE, from noise to data.

From SDS to a Dual Bridge



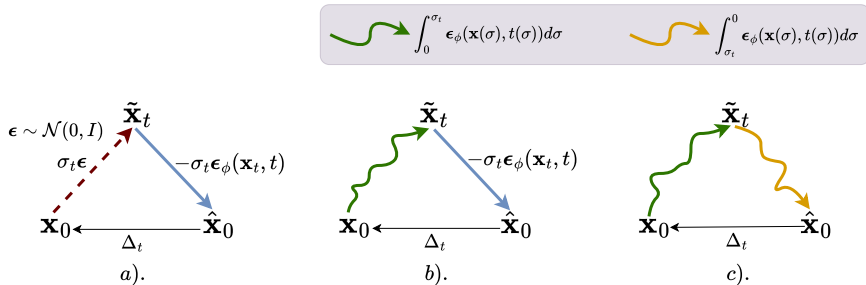
$$\int_0^{\sigma_t} \epsilon_\phi(\mathbf{x}(\sigma), t(\sigma)) d\sigma$$

$$\int_{\sigma_t}^0 \epsilon_\phi(\mathbf{x}(\sigma), t(\sigma)) d\sigma$$



- (a) SDS: Gaussian perturbation forward, single Euler step (posterior mean) back.
- (b) SDI: forward PF-ODE solve, but still a single Euler step back.

From SDS to a Dual Bridge



- (a) SDS: Gaussian perturbation forward, single Euler step (posterior mean) back.
- (b) SDI: forward PF-ODE solve, but still a single Euler step back.

Idea

(c) Replace the posterior mean with a full **reverse PF-ODE** solve:

$$\Delta(\mathbf{x}_0, t) = \mathbf{x}_0 - \Phi_{t \rightarrow 0}^p(\Phi_{0 \rightarrow t}^{q_\tau}(\mathbf{x}_0))$$

The Dual-Bridge Gradient

When $q_{0,\tau} = p_0$, $\Phi_{t \rightarrow 0}^p$ inverts $\Phi_{0 \rightarrow t}^{q_\tau}$ and Δ vanishes for every particle: p_0 is now a fixed point.

Theorem

With $\mathbf{x}_s = \Phi_{0 \rightarrow s}^{q_\tau}(\mathbf{x}_0)$,

$$\Delta(\mathbf{x}_0, t) = \int_0^t \frac{1}{2} g_s^2 \frac{\partial \Phi_{s \rightarrow 0}^p}{\partial \mathbf{x}}(\mathbf{x}_s) (\nabla_{\mathbf{x}} \log q_{s,\tau}(\mathbf{x}_s) - \nabla_{\mathbf{x}} \log p_s(\mathbf{x}_s)) ds,$$

$$\mathbb{E}_t[\Delta(\mathbf{x}_0, t)] = \int_0^1 \frac{1}{2} (1-s) g_s^2 \frac{\partial \Phi_{s \rightarrow 0}^p}{\partial \mathbf{x}}(\mathbf{x}_s) (\nabla_{\mathbf{x}} \log q_{s,\tau}(\mathbf{x}_s) - \nabla_{\mathbf{x}} \log p_s(\mathbf{x}_s)) ds.$$

- $\Delta(\mathbf{x}_0, 1)$ gives exactly the Dual Diffusion Implicit Bridge (DDIB) [15] from $q_{0,\tau}$ to p_0 ; each update $\mathbb{E}_t[\Delta(\mathbf{x}_0, t)]$ moves particles in a direction correlated with DDIB.
- **Cost:** two concatenated PF-ODEs per step, doubling the cost of SDI!

Probability-Flow Distillation (PFD)

Without the Jacobian $\partial \Phi_{s \rightarrow 0}^p / \partial \mathbf{x}$, the gradient requires only the **forward** PF-ODE. PFD drops it:

$$\frac{d\mathbf{x}_0}{d\tau} = \int_0^1 w(t) (\nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t) - \nabla_{\mathbf{x}} \log q_{t,\tau}(\mathbf{x}_t)) dt,$$

$$\mathbf{x}_t = \Phi_{0 \rightarrow t}^{q_\tau}(\mathbf{x}_0)$$

- $w(t) = \frac{1}{2}(1 - t) g_t^2$ (not restrictive; the following theorems are $w(t)$ -agnostic). Approximate $\nabla_{\mathbf{x}} \log q_{t,\tau}$ from an auxiliary model ϵ_ψ , trained online.
- Replace sampling t every step with quadrature on N nodes along *one* forward solve $\Rightarrow$ no Monte Carlo variance in t .

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- Replace sampling t every step with quadrature on N nodes along *one* forward solve $\Rightarrow$ no Monte Carlo variance in t .

NFEs per Step (N nodes)

	SDS	SDI	VSD	PFD	Dual bridge
Evaluations per node	1	2	2	2	$\mathcal{O}(N)$
Total	$\mathcal{O}(N)$	$\mathcal{O}(N)$	$\mathcal{O}(N)$	$\mathcal{O}(N)$	$\mathcal{O}(N^2)$

Probability-Flow Distillation Algorithm

Algorithm Probability-Flow Distillation (PFD)

```
1: Input:  $\{\theta_i\}_i, \{t_k, w_k\}_{k=1}^N, \epsilon_\phi, \epsilon_\psi, \eta$  ▷ particles, nodes, models, lr
2: while not converged do
3:   Update  $\epsilon_\psi$  with DSM on  $g(\theta_i, \omega)$  ▷ tracks  $q_{t,\tau}$ 
4:   for all particles  $i$  do
5:      $\omega \sim \xi, \quad x_0 \leftarrow g(\theta_i, \omega)$ 
6:      $\{x_{t_k}\}_k \leftarrow \Phi_{0 \rightarrow t_k}^{q_\tau}(x_0)$  ▷ forward PF-ODE
7:      $G \leftarrow \sum_k \frac{w_k}{\sigma_{t_k}} (\epsilon_\psi - \epsilon_\phi)(x_{t_k}, t_k)$  ▷ score difference
8:      $\theta_i \leftarrow \theta_i + \eta (\partial x_0 / \partial \theta)^\top G$  ▷ particle update
9:   end for
10: end while
11: Output:  $\{\theta_i\}_i$ 
```

VSD [10] has the same gradient form, but a **stochastic** coupling:

$$\Delta_{\text{VSD}} = w(t) (\nabla_{\mathbf{x}} \log q_{t,\tau}(\mathbf{x}_t) - \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)), \quad \mathbf{x}_t = \mathbf{x}_0 + \sigma_t \epsilon$$

PFD couples $(\mathbf{x}_0, \mathbf{x}_t)$ **deterministically** $\Rightarrow$ its gradient is deterministic up to ω .

Theorem (Small-time regime, $\sigma_t = t$ on $[0, \eta]$)

From a common $q_{0,\tau}$ decaying at infinity, with $\dot{D} := \text{d}D_{\text{KL}}(q_{0,\tau} \| p_0) / \text{d}\tau$,

$$\dot{D}_{\text{VSD}} - \dot{D}_{\text{PFD}} = C_\eta \left(\mathbb{E}_{q_{0,\tau}} \left[\left\| \nabla^2 \log p_0 - \nabla^2 \log q_{0,\tau} \right\|_F^2 \right] + \mathcal{O}(\eta^2) \right)$$

with $C_\eta = \frac{1}{2} \int_0^\eta w(t) t^2 \text{d}t > 0$.

VSD [10] has the same gradient form, but a **stochastic** coupling:

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with $C_\eta = \frac{1}{2} \int_0^\eta w(t) t^2 dt > 0$.

Takeaway

PFD decreases $D_{\text{KL}}(q_{0,\tau} \| p_0)$ at least as fast as VSD for small η , even though VSD is the steepest descent of its own time-averaged KL objective.

Gaussian Case: PFD Is the Fastest

For $p_0 = \mathcal{N}(\boldsymbol{\mu}, \rho^2 \mathbf{I}_d)$ and $\eta = 1$, PFD, VSD, and the dual bridge (DB) all keep the isotropic Gaussian family invariant and admit p_0 as their **unique, globally attracting** fixed point:

$$\frac{d\mathbf{m}_\tau}{d\tau} = -A_M(\mathbf{m}_\tau - \boldsymbol{\mu}), \quad \frac{d\nu_\tau}{d\tau} = (\rho^2 - \nu_\tau^2) B_M(\nu_\tau),$$

for $M \in \{\text{PFD}, \text{VSD}, \text{DB}\}$.

Ordering of the Rates

$$A_{\text{PFD}} = A_{\text{VSD}} > A_{\text{DB}}, \quad B_{\text{PFD}} > B_{\text{VSD}}, B_{\text{DB}}$$

PFD decreases $D_{\text{KL}}(q_{0,\tau} \| p_0)$ strictly faster than DB, and strictly faster than VSD whenever $\nu_\tau \neq \rho$.

Surprisingly, dropping the Jacobian makes PFD converge faster than the dual bridge!

Outline

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Synthetic Targets: 1D Gaussian

Sampling $(g(\theta, \omega) = \theta)$ from $p_0 = \mathcal{N}(2, 0.5^2)$ with 10^5 particles, where Propositions 1, 2, and 6 can be checked directly.

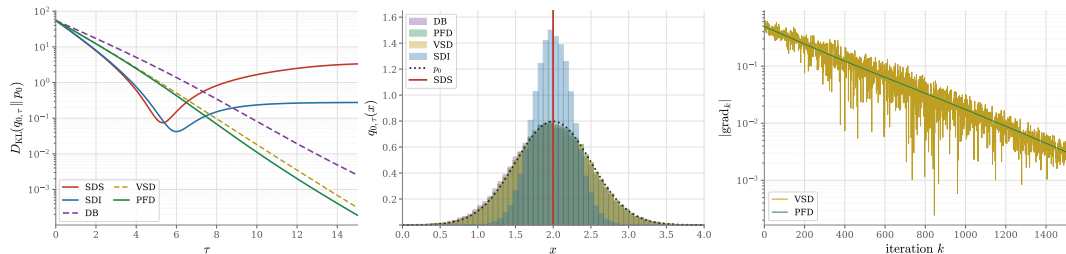


Figure: (left) $D_{KL}(q_{0,\tau} \| p_0)$ vs. optimization time. (middle) Final particle distributions. (right) Gradient magnitude of VSD and PFD for a single particle.

- D_{KL} plateaus for SDS and SDI; keeps decreasing for DB, VSD, PFD, fastest for PFD.
- PFD's gradient is deterministic; VSD's is noisy.

Synthetic Targets: Funnel and Mixture of Gaussians

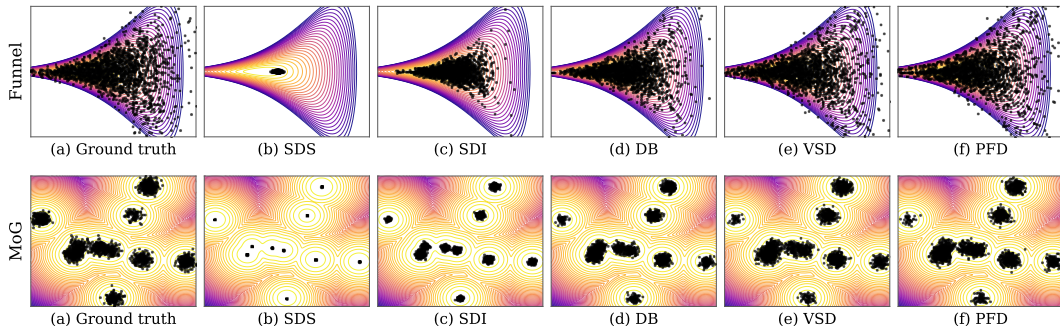


Figure: $2 \cdot 10^3$ particles on Neal's 2D funnel (top) and a 20D MoG with 10 components (bottom, projected onto 2 principal components).

- SDS collapses onto the modes; SDI converges to a contracted p_0 ; DB, VSD, PFD match p_0 .
- The slight contraction of DB comes from discretization error ($N + 1$ ODE solves per step).

CelebA: Four Tasks, One Pretrained Model

A VE diffusion model trained on CelebA [16], used as the prior for four deterministic forward models:

Task	Forward model $g(\theta)$
Image generation	θ
Symmetric generation	$[\theta, \text{flip}(\theta)]$
Inpainting	$M \odot \theta + (\mathbf{1} - M) \odot x^{\text{obs}}$
Vector graphics	diffvg rasterizer [17] on Bézier strokes

What Should Happen

Particles should spread so that $g_{\#}\pi_{\tau}$ gets as close to p_0 as the range of g allows, e.g., inpainting should cover the *plausible completions*, not a single one.

CelebA: Image Generation

Table: FID [18] and CMMD [19] over 10k samples.

Method	FID ↓	CMMD ↓
<i>Reference</i>		
Dataset	2.89	< 0.001
Euler 50	25.54	0.377
Heun 50	24.43	0.368
<i>Distillation</i>		
SDS	236.82	6.531
SDI	108.78	3.289
VSD	16.61	0.106
PFD	16.92	0.106

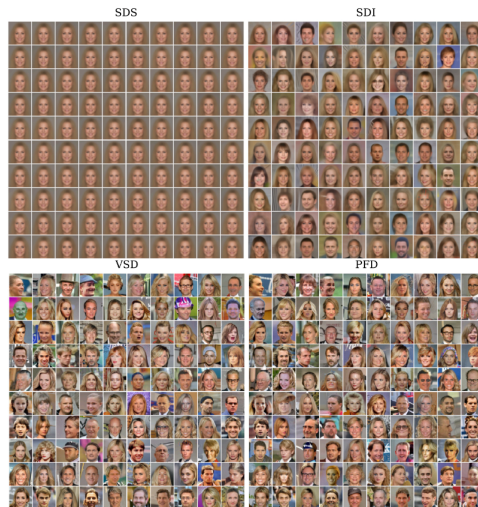


Figure: Uncurated samples: SDS (near-identical), SDI (smooth), VSD and PFD (diverse, detailed).

CelebA: Symmetric Generation and Vector Graphics



Figure: Symmetric generation, $\mathbf{x}_0 = [\theta, \text{flip}(\theta)]$.



Figure: Vector graphics, $\mathbf{x}_0$ is the rasterized render.

CelebA: Inpainting



Figure: Left half masked. First column: initialization; others: 8 uncured particles per method.

Text-to-3D Generation: Setup

- θ : an Instant-NGP [20] volume; g : differentiable renderer; ω : camera pose.
- **Single-particle regime:** $\pi_\tau = \delta_{\theta_\tau}$, so $q_{0,\tau} = g(\theta_\tau, \cdot)_{\#}\xi$ is the distribution of renders over camera poses.
- Matching p_0 asks every view $g(\theta_\tau, \omega)$ to look like a sample from $p_0(\cdot \mid c(\omega)) \Rightarrow$ finer texture and geometry.
- All methods are built on threestudio [21] with the same per-step cost; SDI keeps its DDIM inversion with a negative CFG scale.

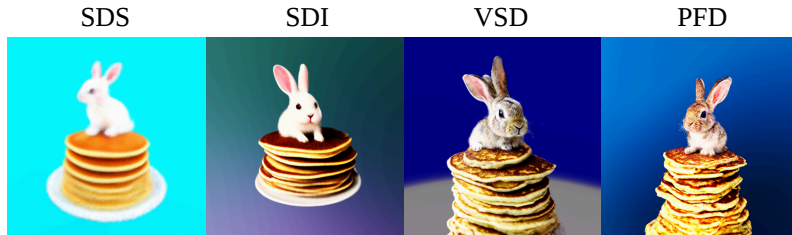
Text-to-3D: Quantitative Results

Table: CLIP R-precision [22] and ImageReward (IR) [23], averaged over 120 views of assets from 22 prompts.

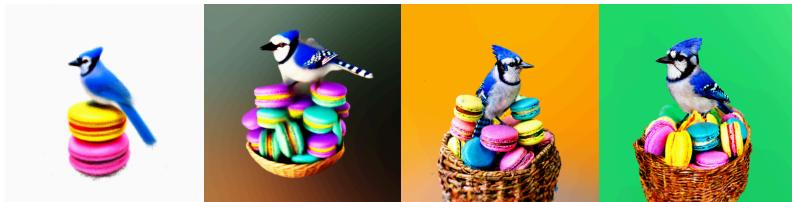
Method	CLIP R-precision $\uparrow$			IR $\uparrow$
	ViT-L/14	ViT-B/16	ViT-B/32	
SDS [4]	0.796	0.820	0.802	-1.150
SDI [9]	0.925	0.917	0.886	-0.257
VSD [10]	0.958	0.955	0.998	0.680
PFD	0.969	0.955	0.998	0.891

- SDS pushes every view to the same high-density image: oversaturated, oversmoothed.
- SDI avoids collapse, but its contraction leaves the views smooth.
- VSD and PFD recover fine texture and geometry; PFD has the highest ImageReward.

Text-to-3D: Qualitative Comparison



"A baby bunny sitting on a stack of pancakes."



"A bluejay standing on a large basket of rainbow macarons."

Figure: Qualitative comparison. More results in Figures 10–12 of [24].

- **SDS** has no dependence on $q_{0,\tau} \Rightarrow$ no entropy term $\Rightarrow$ mode collapse.
- **SDI** restores that dependence via DDIM inversion (an approximate forward PF-ODE) $\Rightarrow$ no collapse, but a strictly contracted spread. The same view explains its negative CFG scale.
- Replacing the posterior mean by a reverse PF-ODE solve makes p_0 a fixed point (dual bridge), at increased computation cost.
- **PFD** drops one Jacobian: only the forward PF-ODE, deterministic gradients, faster than both the dual bridge and VSD (small-time / Gaussian), and competitive or better in practice.
- **Future:** superiority beyond the small-time regime, and developing a multi-particle codebase text-to-3D for *diverse* high-fidelity assets.

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Questions?

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